COMP 3220 Fall 2023 Assignment 2

Due: Oct. 16, 2023, 11:59 PM

Part 1:

Imagine you are tasked with developing a program for a local library. *The program needs to keep track of books, patrons, and library staff interactions*. You can rely on your university/public library experience and common sense when answering the following questions.

a) Create a domain model using appropriate UML diagram for this library software. Identify the main entities, their attributes, and the relationships between them. Be sure to consider the following:

- Books have titles, authors, publication dates, and unique ISBN numbers.
- Patrons have names, contact information, and library card numbers.
- Library staff members have names, staff IDs, and roles (e.g., librarian, clerk).
- Patrons can borrow books, and each book can be borrowed by multiple patrons.
- Library staff members can check books in and out for patrons.

b) Explain your rationale for the design choices you made in your domain model in (a). How did you decide on the entities, attributes, and relationships?

c) Briefly discuss any potential constraints or business rules that your domain model should enforce to ensure the integrity of the library management system.

d) We all know how to borrow book(s) from the library, create a use case called “Borrow books” using fully dressed format to describe the success scenario only (and ignoring all other sections) and draws its SSD as well.

(If you have made any assumptions, please clearly state and write down exactly what your assumptions are when you answer questions.)

Part 2

Question 4 in Assignment 1 describes a streaming service, repeated below.

> *Members join the system by registering online (username (email), password, address, phone no) and including a credit card to pay the membership fees. Most memberships are for one year, but memberships as short as two months are available.*
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> *Throughout the year, the streaming service offers various discounts on its regular membership prices (e.g., two memberships for the price of one for Christmas). It is common for members to pay different amounts for the same length of membership. The service sends out reminder emails to members asking them to renew their memberships one month before their memberships expire.*
>
> *Some members have become angry when asked to renew at a much higher rate than their original membership contract. That is why the organization wants to track the price paid so that the manager can override the regular prices with special prices when members are asked to renew.*
>
> *The system must track these new prices so that renewals can be processed accurately. The manager wants the system to track each time a member visits the site. The system will then identify the heavy users and generate a report so that the manager can ask them to renew their memberships early, perhaps offering them a reduced rate for early renewal. Likewise, the system should identify members who have not visited the site in more than a month so that the manager can email them and attempt to re-interest them in the service.*

a) Model the domain concepts and their relationships. Draw the Domain Model using appropriate UML diagram.

The Domain Model should:

- Be expressed at the problem domain level of abstraction; it should focus on the abstract concepts described above and not include design or implementation classes.
- Include relevant and necessary associations.
- Include multiplicity if appropriate.
- Do not include operations or methods in your conceptual model.

b) You wrote the use case “maintain service usage records” for this streaming service in your Assignment 1. Produce the SSD for the success scenario of this use case.

Use StarUML or other software of your choice to draw appropriate UML diagrams and submit the diagrams as pictures. Please note that you will not get any credit for hand-drawn diagrams.